

ECS 204 — Signals and Systems

Module 3

Fourier Series Representation of Periodic Signals

Section A — Basic Problems

Problem 1

Plot the periodic signal $x(t) = 3 + 2\cos(t)$

Observe:

- DC component
 - Amplitude
 - Fundamental period
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Problem 2

Generate the signal $x(t) = 2\cos(t) + \sin(2t)$

Answer the following:

1. Identify the harmonics.
 2. Is the signal periodic?
 3. Find its fundamental period.
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Problem 3

For $x(t) = \cos(t)$

Determine the exponential Fourier Series coefficients.

Problem 4

Generate $x(t) = \cos(t) + 0.5\cos(3t)$

Compare it with $x(t) = \cos(t)$

How does the third harmonic modify the waveform?

Section B — Intermediate Problems

Problem 5

Verify whether $x(t) = |\sin t|$ satisfies the Dirichlet Conditions.

Problem 6

Generate one period of a square wave and verify whether it satisfies the Dirichlet Conditions.

Problem 7

Approximate a square wave using

- 3 harmonics
- 5 harmonics
- 15 harmonics

Comment on the convergence.

Problem 8

Explain Gibbs Phenomenon using suitable plots.

Problem 9

Plot the amplitude spectrum of $x(t) = 1 + \cos t$

Problem 10

Plot the phase spectrum of $2\cos(t + \frac{\pi}{4})$

Section C — Analytical + Programming

Problem 11

Compare the amplitude spectra of

$$x_1(t) = \cos t \text{ and } x_2(t) = \cos t + 0.5\cos 3t$$

Problem 12

Verify the Linearity Property of Fourier Series using Python.

Problem 13

Verify the Time Shifting Property.

Problem 14

Verify the Differentiation Property.

Problem 15

Verify Parseval's Theorem numerically.

Challenge Problems

Challenge 1

Approximate a triangular wave using Fourier Series.

Challenge 2

Approximate a sawtooth wave.

Challenge 3

Write a Python function

`fourier_series(coefficients, t)`

that reconstructs any periodic signal from its Fourier coefficients.

Challenge 4

Investigate how the waveform changes when only even harmonics are retained.

Challenge 5

Investigate how the waveform changes when only odd harmonics are retained.

Challenge 6

Approximate a square wave using

- 5 terms
- 15 terms
- 25 terms
- 75 terms

Plot all four approximations in one figure and compare them.